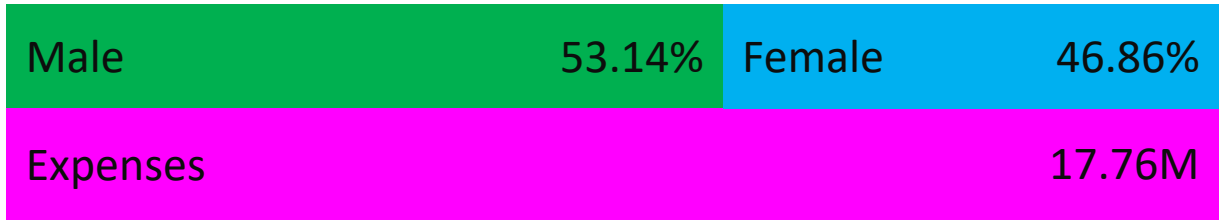
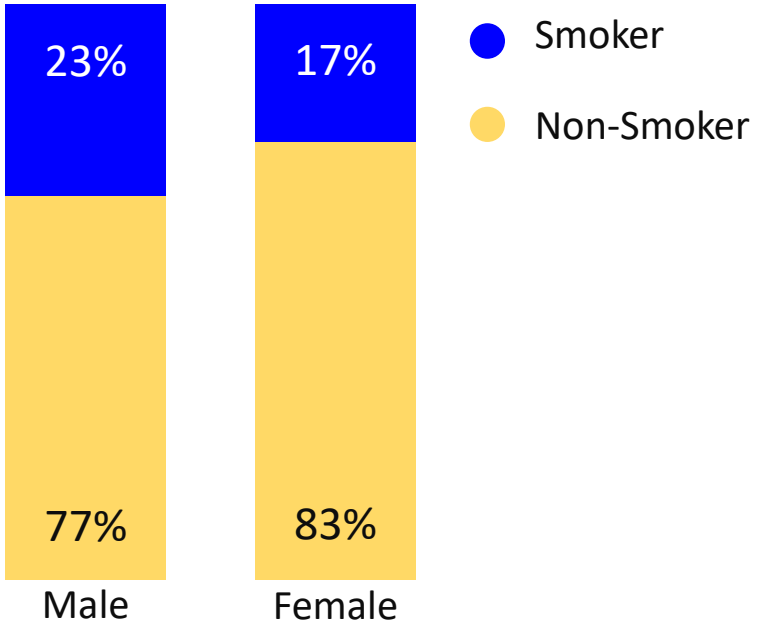
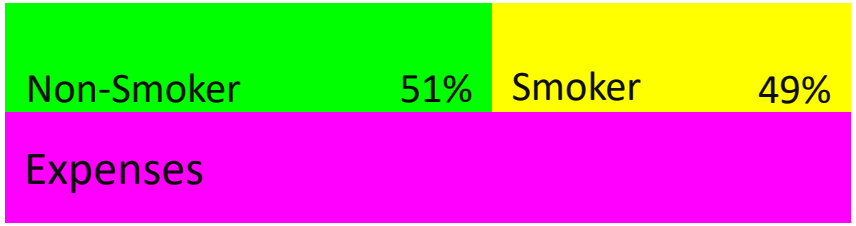
Insurance Premium Prediction

Analysis Report

Expenses by Gender (Male & Female):



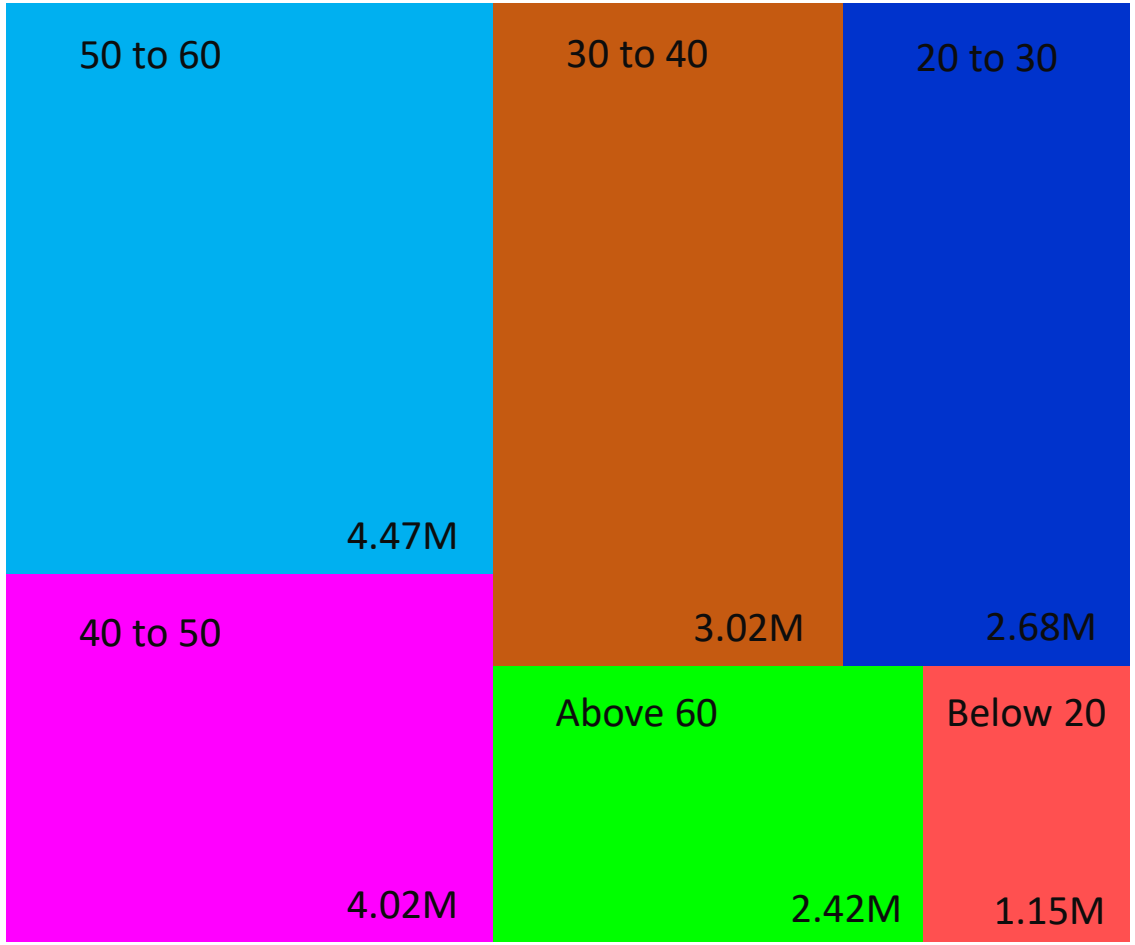




Insight:

- Male are expenses more i.e., 53.14% compare to Female i.e., 46.86%

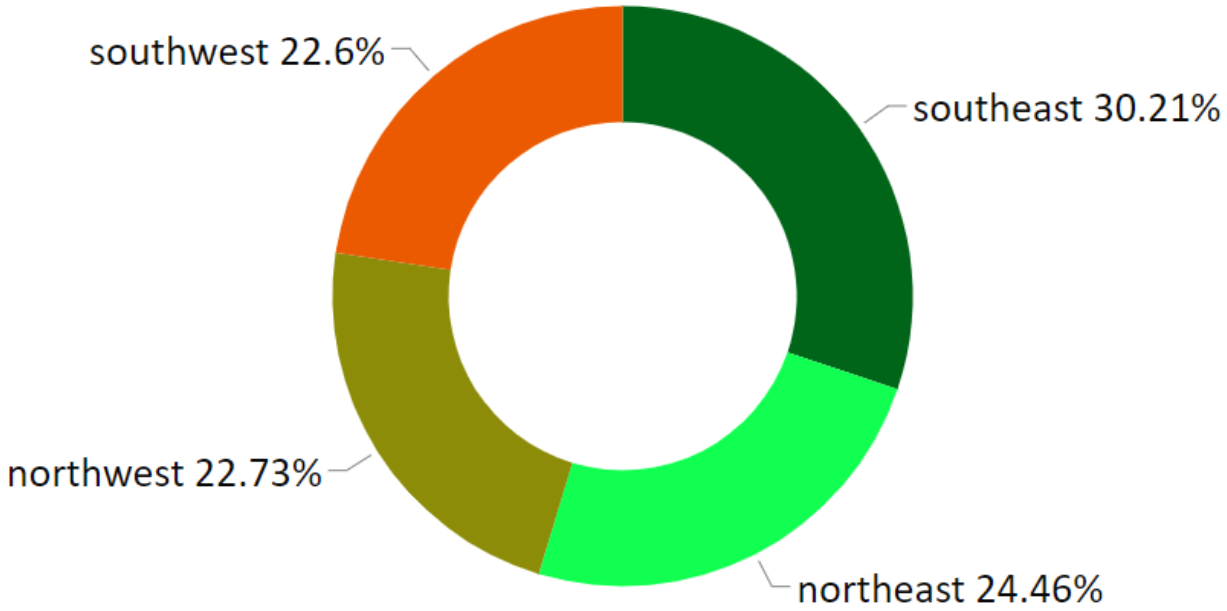
Expenses by Age Group:



Insight:

- In 50-60 years age group expense more compare to 30-40 & 20-30 years age group.

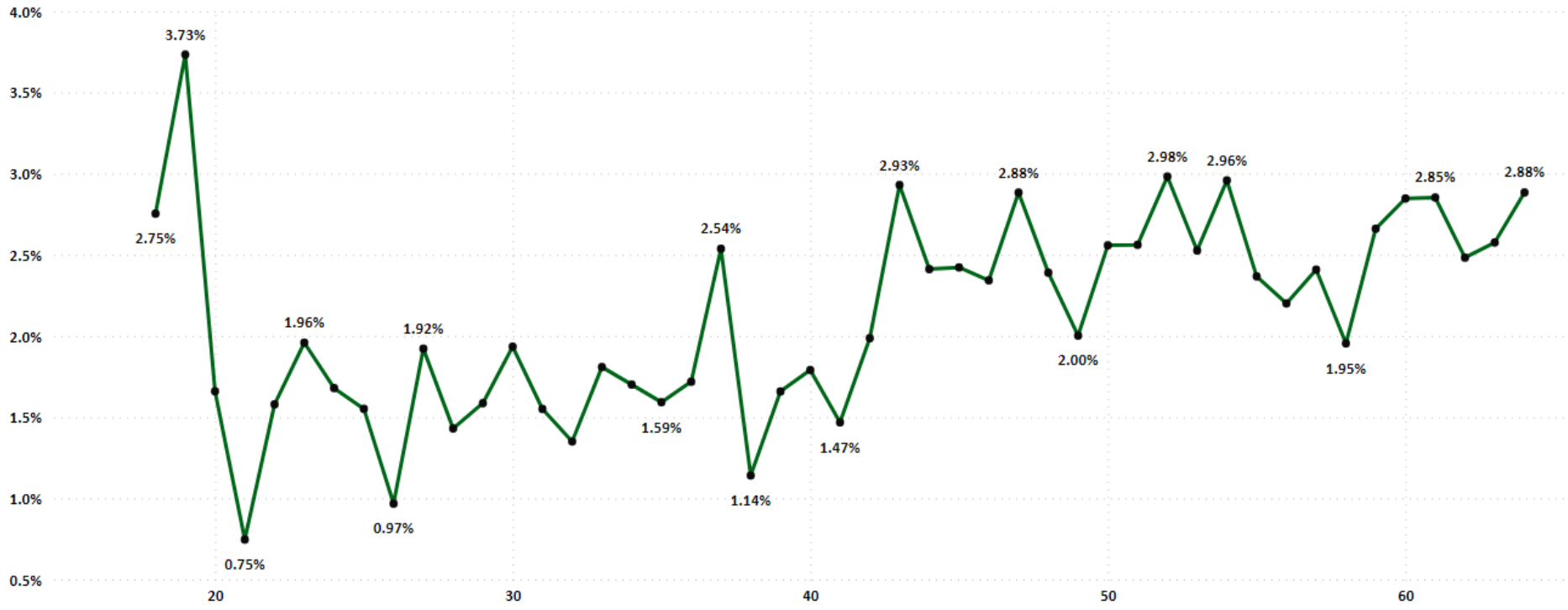
Expenses by Region:



Insight:

- Almost all regions people expenses equally, but southeast compare to more, i.e., 30.21%.

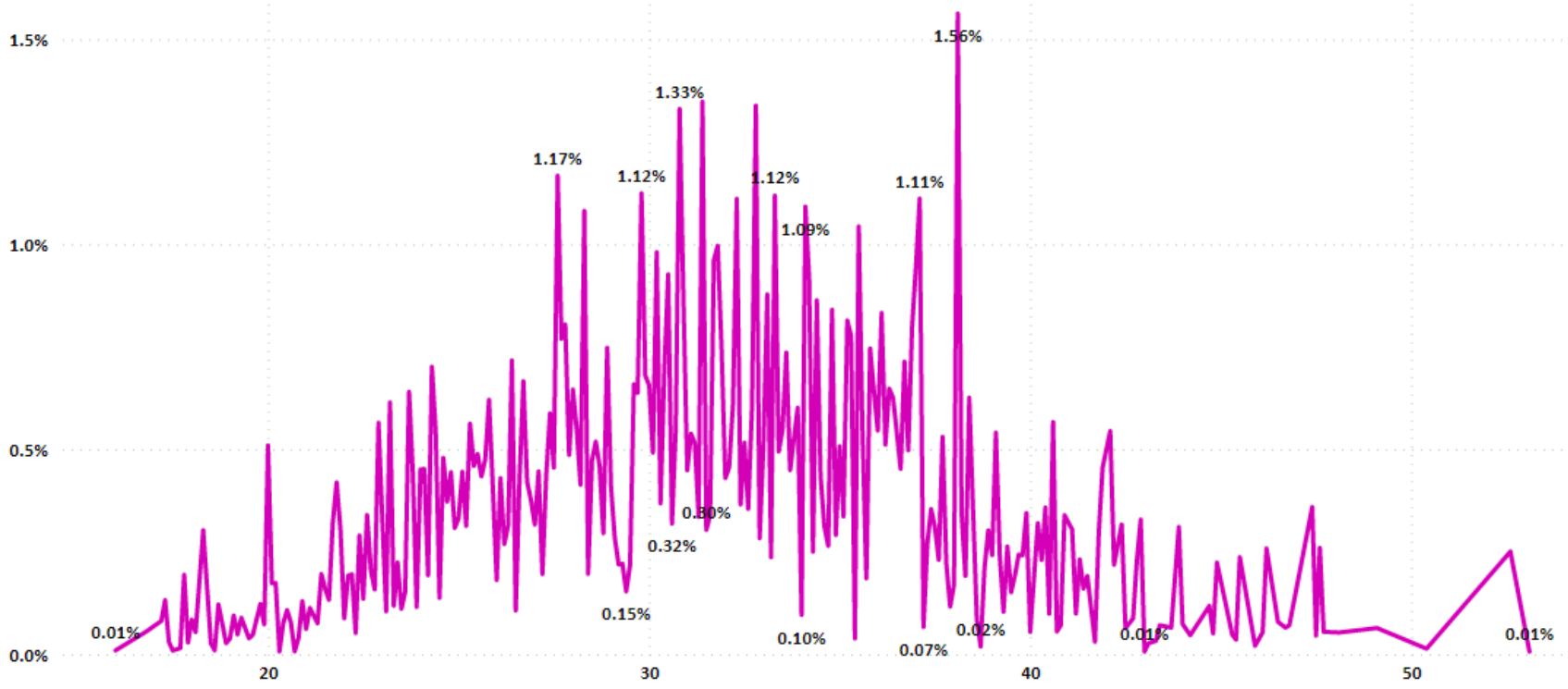
Expenses by Age:



Insight:

- Age wise expenses ratio is flatulate, not an increasing order over age.

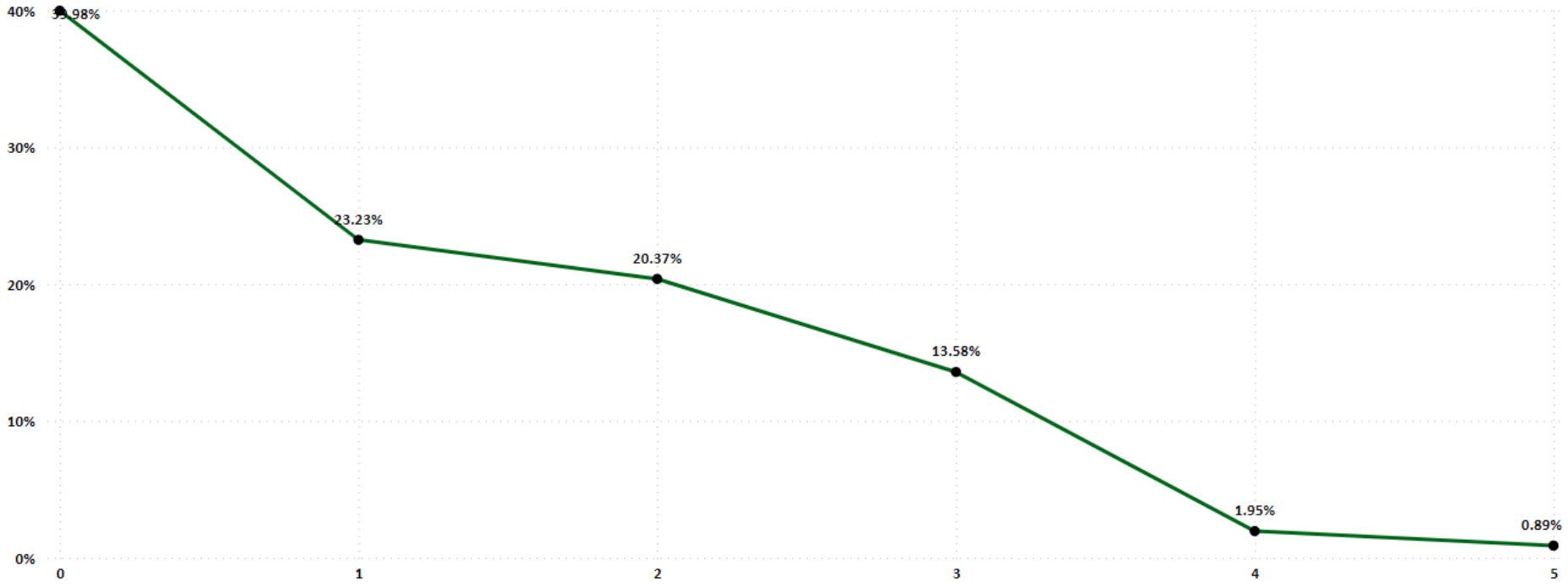
Expenses by BMI:



Insight:

- BMI wise expenses ratio is flatulate, not an increasing order over bmi.

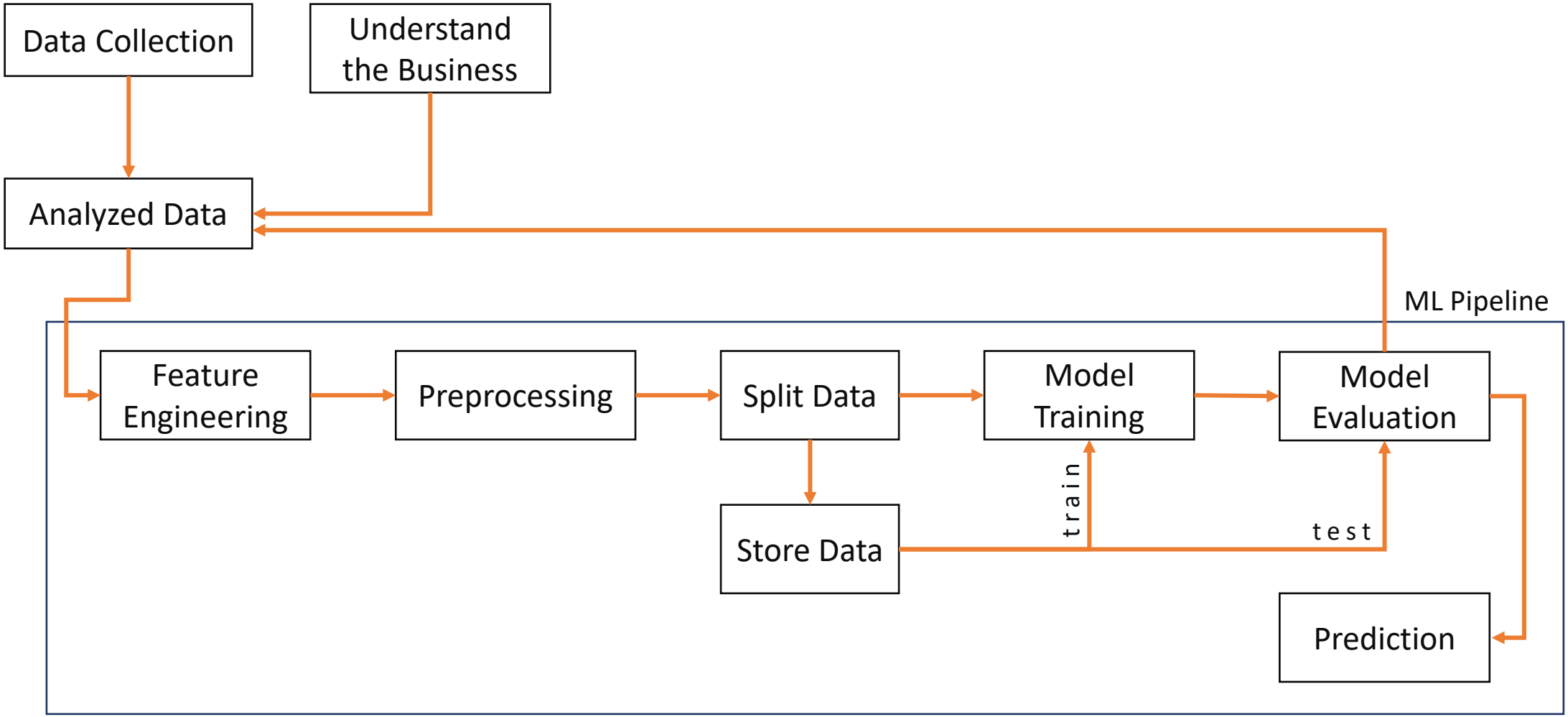
Expenses by Children:



Insight:

- When number of children of a person increase then expenses also decrease.

Machine Learning Pipeline:

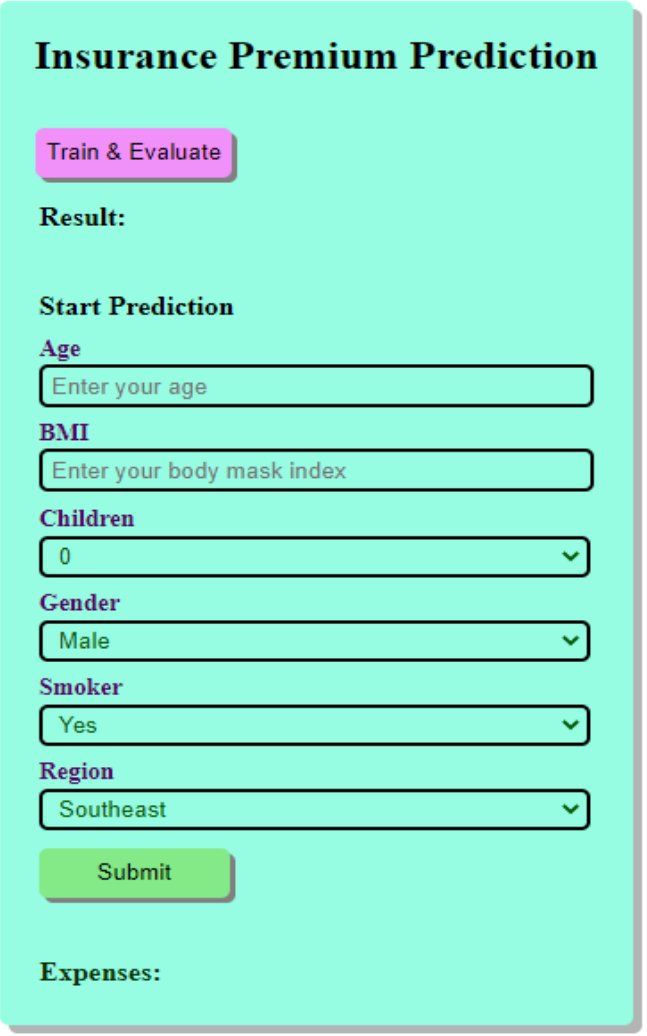


Generalized Machine Learning Model:

- In this project there is no such more linear relationship between independent features and dependent feature, so we use there ML algorithms to build the model, i.e.,
  - RandomForestRegressor,
  - SupportVectorRegressor (SVR), and
  - GradientBoostingRegressor.

- Here, we select GradientBoostingRegressor algorithm based on performance matrix. [Matrix Link](#).

- After that create a web app for predict the expenses based on new test data.



-by Dibyendu Biswas